

right or even U-turn as per directions sensed by the application.

The bike has numerous inbuilt sensors including accelerometers, gyroscopes, magnetometer, GPS receiver, speed sensors and blind-spot sensors. All these sensors enable it to predict its motion.

Valour has integrated ultrasonic sensors that detect a safe-zone radius. In case any object comes too close and enters the safe zone, the ultrasonic sensors pick it up and handles start vibrating, alarming the rider of a potential chance of collision. The sensors consider 1.2m (4 feet) along the sides and 3m (10 feet) behind the bike as the safe zone threshold. These blindspot sensors add X-factor to the bike and justify the definition of smart bikes. Valour can also evaluate the road quality through its 9-axi accelerometer.

In addition to this, there is an anti-theft alarm system. In case the bicycle has gone missing, the mobile application tracks the in-built signature of the bike. As soon as it is located, the application provides the location and timestamp where the bike has been tracked. The owner can also share the details to the nearest police

station immediately, details of which are shared by the application as well!

Vanhawks lets its customers choose the frame size, colour and speed options before buying. Based on the road types, the bike can be single-speed or dynamic-speed. Its body is made of light yet rugged carbon-fibre and weighs 9-13kg. Vanhawks Valour is priced at \$1500.



VanMoof SmartBike

VanMoof SmartBike

VanMoof has introduced a smart bike that unlocks with your touch. Well, there is no fingerprint sensor *per se*. To unlock the bike, you should either use the mobile application provided by VanMoof or turn your phone's Bluetooth 'on' and just touch the display of the bike. The sensors in the bike will detect the phone's Bluetooth, identify

Vanhawks Valour bicycle with handlebar highlighted



it and let you access the bike.

The mobile application has in-built intelligence. Based on your ride metrics and patterns, daily schedule and other data, it can start giving recommendations and predictions.

The anti-theft system is also quite capable. The first layer of protection comes from the hardware. Bike parts that are more likely to be stolen are